



UNIVERSITY OF KARACHI

Department of Computer Science

KuCampusBite

Campus Food Ordering and Delivery Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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Preamble

This Software Requirements Specification (SRS) presents the requirements analysis for KuCampusBite, a web-based food ordering and delivery platform for the University of Karachi. It was prepared as part of the Software Project Management course project and is intended to give the instructor, project team, and other stakeholders a shared, written understanding of what the system must do.

The document covers the project purpose and scope, an overall description of the product, the functional, non-functional, and interface requirements, and supporting system models, including an entity relationship diagram and a use case diagram. It should be read alongside the Business Case, Project Charter, and Project Plan, which provide the wider context for this work.

Abstract

Students at the University of Karachi currently face long queues, crowded canteens, and limited access to food from distant departments, which wastes time and reduces satisfaction with campus dining. This document specifies the requirements for KuCampusBite, a centralized web-based platform that addresses these problems by bringing all campus canteens into a single ordering interface. The objective is to let students and teachers browse menus, place orders, and receive food at their campus location free of delivery charge, while giving canteen staff a simpler way to handle incoming demand and reduce crowding.

The system will be delivered as a mobile-responsive website built with HTML, CSS, and JavaScript on the frontend and Node.js with Express.js on the backend, using MongoDB Atlas for data storage. Orders will be forwarded to canteen staff through WhatsApp, and menu availability will be maintained through a Google Form linked to a Google Sheet, keeping the solution lightweight and inexpensive to operate. The first release will operate on a cash-on-delivery basis and serve on-campus locations only; payment gateways, university system integration, and live order tracking are excluded, while order scheduling, WhatsApp status updates, and ratings are planned enhancements. The expected outcome is faster, more organized food access for the university community, shorter waiting times, and a foundation that can be extended in later versions.

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1 Introduction

This project proposes the development of a centralized, web-based food ordering and delivery platform for the University of Karachi. The platform will provide students with a convenient and cost-effective alternative to physically traveling between departments and distant canteens. It aims to save time, reduce transportation costs, and simplify the process of ordering food, especially during peak hours.

The system will support multiple canteens with varying menus, pricing, delivery charges, and policies, while ensuring a lightweight, secure, and user-friendly experience suitable for campus use.

It will also assist canteen staff in managing orders efficiently, helping to reduce crowding and long waiting times. Additionally, the platform will offer a free delivery service to make food access more affordable and accessible for all users.

1.1 Purpose

The purpose of this project is to improve the accessibility, speed, and convenience of food services within the University of Karachi.

At present, students face challenges such as long queues, crowded canteens, and limited access to food from other departments due to distance. These issues result in wasted time and reduced overall satisfaction with campus dining options.

The proposed system will address these challenges by enabling users to place orders online from any canteen on campus and have them delivered free of charge to their location.

For canteen staff, it will provide a structured order management system to handle incoming requests efficiently, thereby reducing operational strain and overcrowding. Ultimately, the platform aims to create a faster, more organized, and more accessible food service environment for the entire university community.

1.2 Scope

The proposed system is a centralized, web-based food ordering and delivery platform for the University of Karachi, designed to streamline the process of ordering meals from multiple on-campus canteens. The platform will prioritize accessibility, ease of use, and operational efficiency for both students/teachers and canteen staff.

1.2.1 Included Features

1. **Meal Pre-Ordering, Scheduling, and Reservation** – Users will be able to place orders in advance, schedule a delivery time, and reserve meals to ensure availability during peak hours.
2. **Secure User Login and Basic Profile Management of Canteens** – Each canteen will have a secure login to manage its own profile, including menu details, operating hours, and delivery policies.
3. **Canteen Dashboard for Managing Orders and Menus** – A dedicated interface for canteen staff to view, update, and process incoming orders efficiently, as well as modify menu items when needed.
4. **Mobile-Responsive Design for Student Accessibility** – The platform will be optimized for mobile devices to allow students to place orders easily from anywhere on campus.
5. **Initial Training and Onboarding of Canteen Staff** – Basic training will be provided to canteen staff to ensure they can effectively use the system for order management and customer service.

1.2.2 Excluded Features

1. **Payment Gateway Integration** – The initial version will not include online payment options; all transactions will be processed via cash on delivery.
2. **Integration with Student ID Cards or University Systems** – The platform will operate independently without linking to existing university databases or student ID systems.
3. **Off-Campus Deliveries** – The delivery service will be limited strictly to locations within the university campus.
4. **Multilingual Interface** – The platform will initially be available in English only, with potential for additional languages in future updates.
5. **Mobile Application Development** – The system will be delivered strictly as a responsive website, with no downloadable mobile app in the initial release.
6. **Real-time Tracking** – The initial release will not include a live update feature showing the current status of a food order; this may be introduced in future versions. Only an estimated delivery time will be shown.

1.3 Overview

The proposed system is a centralized, web-based platform designed to simplify and improve the process of ordering and delivering food within the University of Karachi. The platform will cater to students, offering a convenient alternative to traveling between departments and distant canteens. By integrating multiple canteens into a single interface, the system will allow users to view menus, place orders, and schedule deliveries without physically visiting the canteens.

The platform will be mobile-responsive to ensure accessibility on smartphones, tablets, and desktop computers. Canteen staff will be provided with a dedicated dashboard for managing menus, processing orders, and tracking deliveries.

While the initial version will not include real-time order tracking, users will be shown an estimated delivery time to set expectations. The system will operate on a cash-on-delivery model and will only cover deliveries within the university campus.

Ultimately, the platform aims to save time and travel cost, reduce congestion in canteens, improve operational efficiency for staff, and make food access quicker and more convenient for the university community.

Technical terms and acronyms used in this document are defined in the Glossary at the end of the document, and the supporting project documents are listed under References.

2 Overall Description

2.1 Product Perspective

This project is a new, centralized web platform that connects all university canteens. It replaces the current manual system of physically visiting canteens and waiting in queues.

The system will work as a central hub where students can browse menus, order food, and get it delivered anywhere on campus.

2.2 Product Functions

- Show menus from all canteens.
- Allow users to place and schedule orders.
- Track orders in real time.
- Rate food and delivery service.
- Enable canteens to manage menus and stock.
- Allow canteen staff to accept and update orders.
- Provide delivery management for campus riders.

2.3 User Classes and Characteristics

- **Students & Staff (Users):** Order and track food; rate service.
- **Canteen Staff:** Manage menus, process orders, update availability.
- **Delivery Team:** Pick up and deliver food.
- **University Admin:** Monitor performance and manage operations.

2.4 Operating Environment

- Web-based application accessible on computers, tablets, and mobile phones.
- Requires internet access (Wi-Fi or mobile data).
- Works on modern browsers (Chrome, Firefox, Edge, Safari).

2.5 Design and Implementation Constraints

- Must be accessible only within the university network for security.
- Real-time updates needed for menus and order status.
- Budget and development time constraints.
- Needs simple interface for all user types.

2.6 User Documentation

- **User Guide:** Step-by-step instructions for placing orders, tracking, and rating service.
- **Admin Guide:** Instructions for managing menus, orders, and delivery schedules.
- **Training Materials:** Simple video tutorials or physical training for new users and FAQs.

3 Specific Requirements

3.1 Functional Requirements

The functional requirements describe the main operations that the KuCampusBite system will provide to its users, canteen staff, and administrators. The main functional requirements are summarized in Table 3.1.

Table 3.1: Functional Requirements

Requirement	Description
User Login	Users shall be able to log in using a valid email address and password.
Admin Login	Administrators shall be able to log in using a valid email address and password.
Guest Access	Users shall be able to place orders without creating an account.
Menu Browsing	The system shall display menus for registered canteens, including item name, description, price, and availability status.
Search	Users shall be able to search for food items by name.
Cart Management	Users shall be able to add items to the cart, change quantities, and remove items before checkout.
Order Placement	Users shall be able to confirm and submit orders from the cart.
Order Notification	Order details shall be sent to the respective canteen staff through WhatsApp.
Menu Availability Updates	Canteen staff shall be able to update item availability through the designated Google Form linked with Google Sheets.
Order Management	Administrators shall be able to view all orders placed through the system.
Canteen Management	Administrators shall be able to add, edit, or remove canteen profiles.
Menu Management	Administrators shall be able to add, edit, or remove menu items and update their availability.
Data Storage	The system shall store user, canteen, menu, and order information in a database.

In addition to the requirements listed in Table 3.1, the following features are planned for later releases and are summarized in Table 3.2.

Table 3.2: Planned Features

Planned Feature	Description
Order Scheduling	Users may be able to schedule orders for a future time slot.
WhatsApp Order Status Updates	Users may receive order status notifications via WhatsApp based on canteen staff updates.
Feedback & Ratings	Users may submit feedback or rate orders and canteens.

3.2 Non-Functional Requirements

The non-functional requirements define the expected quality and performance of the KuCampusBite system. The requirements are summarized in Table 3.3.

Table 3.3: Non-Functional Requirements

Category	Requirement
Performance	The menu page should load within 3–5 seconds on a stable internet connection.
Performance	Food-item search results should be displayed within 3 seconds.
Availability	The system should be available at least 90% of the time during testing and demonstration.
Usability	A first-time user should be able to place an order without external guidance within 5 minutes.
Security	User passwords should be stored in hashed form in the database.
Security	Basic input validation should prevent at least 90% of common invalid inputs.
Maintainability	At least 70% of the core functions should be updateable without affecting other modules.
Compatibility	The system should be accessible through at least Chrome and Firefox on both mobile and desktop devices.
Data Accuracy	Menu items and prices displayed by the system should match the database records with 100% accuracy during testing.

3.3 Interface Requirements

The interface requirements describe how users and system components will interact with the KuCampusBite platform. These requirements are summarized in Table 3.4.

Table 3.4: Interface Requirements

Interface Type	Requirement
User Interface	The system shall provide a responsive interface for mobile, tablet, and desktop screens.
User Interface	Menu items shall be displayed as image cards containing the item name, price, and an “Add to Cart” button.
User Interface	The interface shall provide search functionality and category filters for browsing menus.
User Interface	Users shall be able to view their cart contents and total price before confirming an order.
Hardware Interface	The system shall support standard smartphones and computers with internet connectivity.
Hardware Interface	No specialized hardware shall be required for users or canteen staff.
Hardware Interface	Canteen staff shall receive orders through WhatsApp on their smartphones.
Software Interface	The frontend shall use HTML, CSS, and JavaScript.
Software Interface	The backend shall use Node.js with the Express.js framework.
Software Interface	MongoDB Atlas shall be used as the database service.
Software Interface	The system shall integrate with the WhatsApp API for order notifications.
Communications Interface	Communication between the frontend and backend shall use HTTPS and RESTful APIs.
Communications Interface	Order notifications shall be sent to canteen staff through the WhatsApp API.
Communications Interface	JSON shall be used for API requests and responses.
Communications Interface	A stable internet connection shall be required for reliable operation.

4 System Models

4.1 Entity Relationship Diagram

The Entity Relationship Diagram represents the main entities involved in the KuCampusBite system and shows how these entities are related to each other. It provides a visual representation of the data structure used by the system. The relationships between the core entities are illustrated in Figure 4.1.

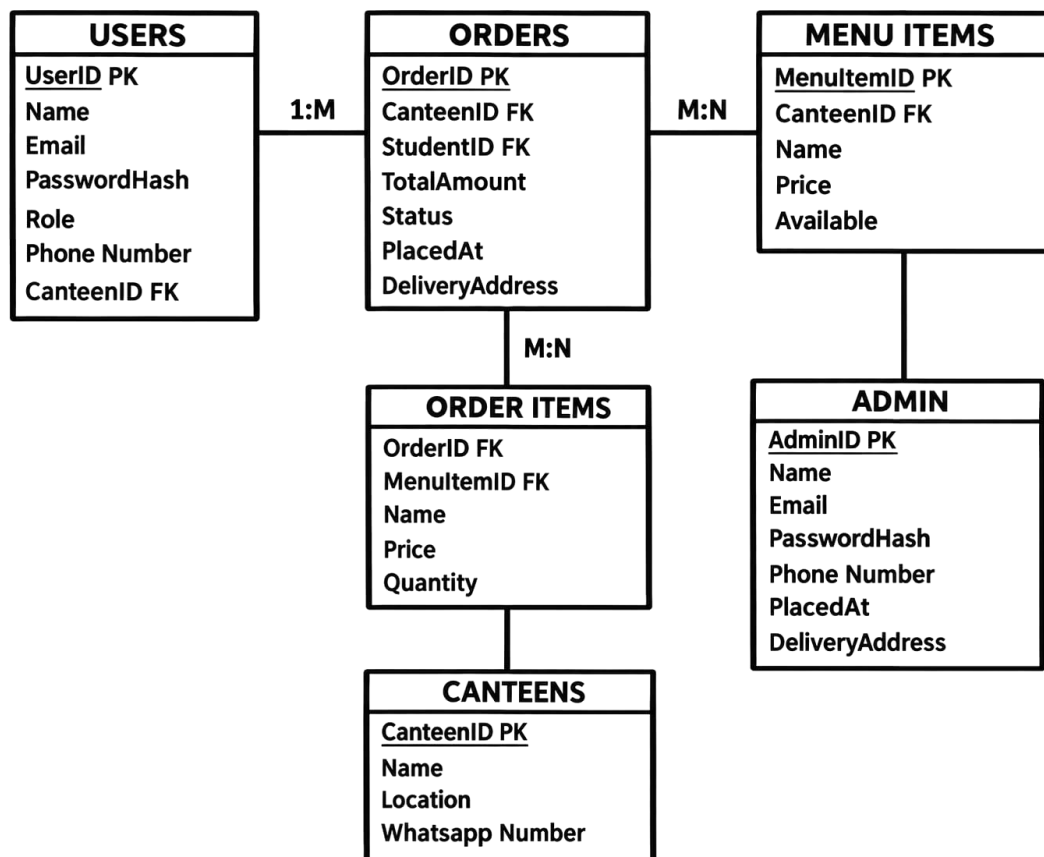


Figure 4.1: Entity Relationship Diagram

4.2 Use Case Diagram

The Use Case Diagram presents the main interactions between the different users and the KuCampusBite system. It shows the activities that can be performed by users such as students or staff, canteen staff, delivery team members, and university administrators. The main system interactions are illustrated in Figure 4.2.

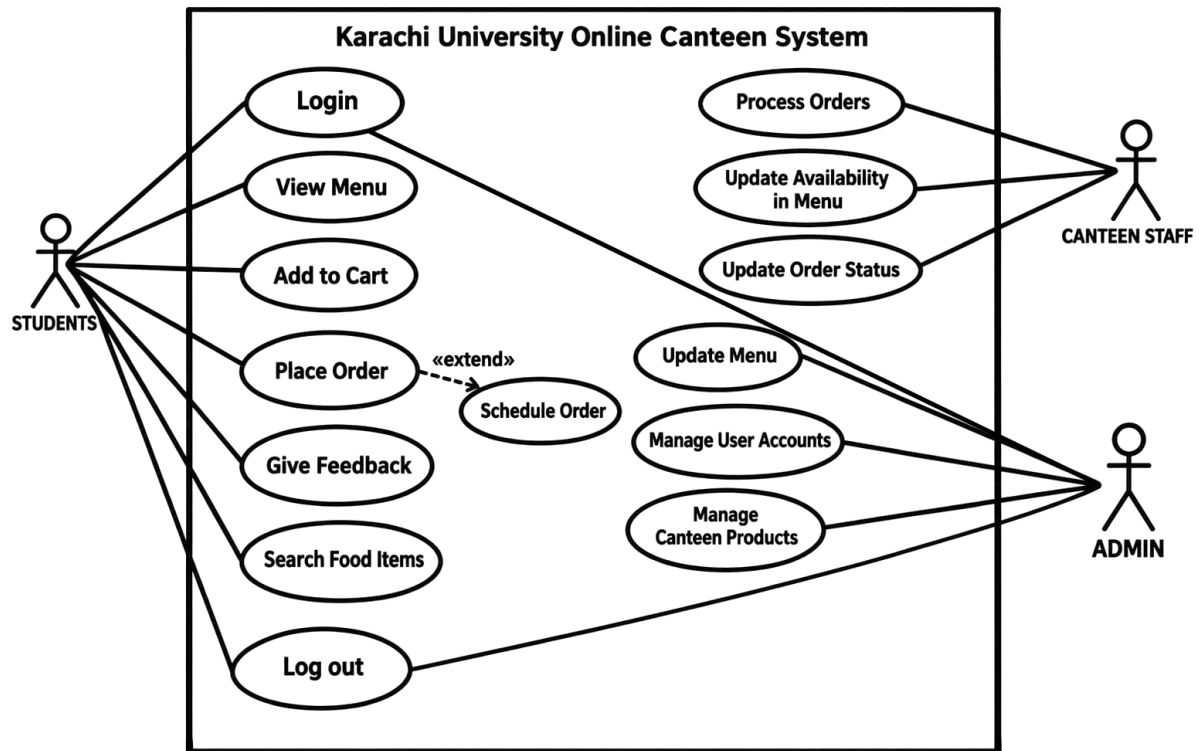


Figure 4.2: Use Case Diagram

Glossary

Term	Definition
Admin	A user responsible for managing canteens, menus, orders, and other system operations.
Canteen Staff	Staff members who manage menus and process food orders received from users.
Delivery Service	The process of delivering prepared food from a canteen to a user's location within the University of Karachi.
Guest User	A user who can place an order without creating or logging into an account.
HTTPS	A secure communication protocol used to protect data exchanged between the user's browser and the web platform.
JSON	A lightweight data format used for sending and receiving information between the frontend and backend.
MongoDB Atlas	A cloud-hosted database service used by the platform to store system data.
Platform	The centralized web-based food ordering and delivery system developed for the University of Karachi.
REST API	An interface that allows the frontend and backend to communicate using web-based requests.
UI	User Interface; the part of the system through which users interact with the platform.
UoK	University of Karachi.
User	A student or staff member who uses the platform to browse food items and place orders.
WhatsApp API	An interface used to send order-related notifications to canteen staff through WhatsApp.

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